

A Comparative Analysis of Classification Algorithms for Driving Style Prediction

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The full Python code for this project is available at: <https://github.com/benedict8005/Machine-Learning-Repo.git>

Abstract

This study investigates the use of machine learning classification techniques to predict driver behavior using multi-journey vehicle sensor datasets from two vehicles (Opel Corsa and Peugeot 207). The objective is to compare the performance of Logistic Regression (LR) and Support Vector Machines (SVM) on the binary classification task of distinguishing Even Pace Style and Aggressive Style driving behaviors. The data set, sourced from Kaggle, contains 14 sensor features from two different vehicles (Opel Corsa and Peugeot 207). A significant class imbalance was observed, which was addressed using the Synthetic Minority Over-sampling Technique (SMOTE). The methodology involved data reprocessing, including feature selection using Recursive Feature Elimination (RFE), followed by Z-score normalization. The models were trained in an 80% split of one Opel Corsa journey and then evaluated in four test phases. Hyperparameter tuning using GridSearchCV to optimize for the F1-score was essential. Although both models showed a performance drop on data from a different vehicle (Peugeot), the SVM consistently provided better recall, demonstrating superior (though limited) generalization for this task.

1 Introduction

Modern vehicles collect a large amount of sensor data that can be leveraged to classify driving behavior in real time. Machine learning offers a robust framework for analyzing these data and identifying patterns related to safe or unsafe driving, offering opportunities like driver safety monitoring and insurance risk assessment. This project investigates Logistic Regression and Simple Vector Machines classification models. Similar-type study, H. Shickel et al. compares SVM and logistic regression for predicting ICU patient mortality, demonstrating the viability of this approach[1]

The primary goal is to compare the performance of two distinct classification algorithms: Logistic Regression (LR), a linear probabilistic model, and a Support Vector Machine (SVM), a non-linear model. We aim to determine not only which model performs best on data from the source vehicle but also how well these models generalize to data from a different journey and, more challenging, from a different vehicle model entirely. The initial structure of the machine learning pipeline was established after brainstorming foundational ideas and processes with an AI model [2].

2 Methodology

2.1 Dataset Description

The raw dataset consists of four CSV files, highlighting two journeys each for two vehicles: an Opel Corsa and a Peugeot 207. The data includes 14 sensor features and three potential classification targets. For this

analysis, the target variable 'drivingStyle' was selected, which classifies behavior into 'EvenPaceStyle' (class 0) and 'AggressiveStyle' (class 1).

A preliminary analysis revealed a significant class imbalance. The Opel Corsa datasets showed a majority class (EvenPace) distribution of approximately 80-82%, while the Peugeot 207 datasets were far more imbalanced, with the majority class comprising 94-96% of the data. Due to its more balanced distribution and lack of missing values, the `opel_corsa_01` file was chosen as the primary dataset for training and initial testing.

2.2 Data Preprocessing and Feature Engineering

The preprocessing pipeline consisted of several key stages. First, any rows containing missing NA values were dropped. The categorical 'drivingStyle' labels were then mapped to numeric values (0 for 'EvenPaceStyle', 1 for 'AggressiveStyle').

The `opel_corsa_01` data was then split into an 80% training set and a 20% test set (Test 1), using stratification to maintain the class distribution in both sets.

Next, Z-score normalization was applied using Scikit-learn's `StandardScaler`. The scaler was **fit only on the training data** to learn its mean and standard deviation. This fitted scaler was then used to **transform** the training set, the hold-out test set, and all subsequent test datasets (Opel 2, Peugeot 1 & 2).

After scaling, feature selection was performed using Recursive Feature Elimination (RFE) with a 3-fold cross-validation. This backward propagation method was configured to select the best 8 features, which were found to be: *AltitudeVariation*, *VehicleSpeedAverage*, *VehicleSpeedVariation*, *ManifoldAbsolutePressure*, *EngineRPM*, *FuelConsumptionAverage*, *VehicleSpeedInstantaneous*, and *MassAirFlow*. These 8 features were used for all subsequent steps.

2.3 Handling Class Imbalance

The training data was highly imbalanced (82% vs 18%). To prevent the models from simply predicting the majority class, the Synthetic Minority Over-sampling Technique (SMOTE) was applied [3]. SMOTE was fit only to the 8-feature, scaled training set, creating a new, balanced training set with a 50/50 class distribution. The models were trained on this resampled data, while all test sets remained in their original, imbalanced state to provide a realistic evaluation.

2.4 Classifier Models and Tuning

Two classifiers were trained on the SMOTEd data. To find the optimal hyperparameters for each, `GridSearchCV` was used with 3-fold cross-validation, optimizing for the 'f1' score to prioritize performance on the 'AggressiveStyle' minority class [5].

Hyperparameter tuning is a critical step for optimizing model complexity. The `C` parameter, tuned for both models, controls the regularization strength; a smaller `C` applies stronger regularization to prevent overfitting. For the SVM, the `kernel` parameter was also tested. The 'linear' kernel attempts to find a simple hyperplane, while the 'rbf' (Radial Basis Function) kernel can capture complex, non-linear relationships, which is what was found to be optimal in this case [4].

1. **Logistic Regression (LR):** A 'liblinear' solver with max iterator of 1000 was used. The `C` parameter (inverse of regularization strength) was tuned over [0.01, 0.1, 1, 10, 100] and reran with other variables until **C=0.6** was found to be optimal.
2. **Support Vector Machine (SVM):** An SVC with `probability=True` was used to enable ROC-AUC scoring. The grid searched 'C' values [10, 1000, 10000], `gamma = "scale"`, and `kernel = ["rbf", "linear"]`. The best parameters were found to be `C=5000`, `gamma="scale"`, and `kernel="rbf"`.

2.5 Experimental Setup

The two best-tuned models were evaluated across four test phases. The classification threshold was adjusted to bias the model towards identifying the 'AggressiveStyle' class, improving recall at the cost of precision for both models. 0.25 for SVM and 0.45 for LR were considered the best fit.

- **Test 1 (Hold-out):** Models were evaluated on the 20% hold-out portion of the `opel_corsa_01` data.
- **Test 2 (New Journey):** Models were evaluated on the full, preprocessed `opel_corsa_02` dataset.
- **Test 3 & 4 (New Vehicle):** Models were evaluated on the preprocessed `peugeot_207_01` and `peugeot_207_02` datasets.

Performance was measured using Accuracy, AUC, and F1-Score, Specificity, and False Positive Rate (FPR) for the 'Aggressive' class.

3 Results and Discussion

The performance of the optimized Logistic Regression (LR) and Support Vector Machine (SVM) models across all four test phases is summarized in Table 1.

Table 1: Final Results Table for all Test Phases.

Model	Test Set	Accuracy	AUC	Prec. (Aggr.)	Recall (Aggr.)	F1 (Aggr.)	Spec. (Even)	FPR (Even)
LR	Test 1 (Opel Hold-out)	0.585	0.743	0.282	0.825	0.421	0.532	0.468
SVM	Test 1 (Opel Hold-out)	0.783	0.885	0.450	0.848	0.588	0.769	0.231
LR	Test 2 (Opel New Journey)	0.629	0.724	0.295	0.642	0.404	0.626	0.375
SVM	Test 2 (Opel New Journey)	0.700	0.684	0.326	0.498	0.394	0.749	0.251
LR	Test 3 (Peugeot J1)	0.666	0.571	0.065	0.348	0.109	0.686	0.314
SVM	Test 3 (Peugeot J1)	0.667	0.609	0.077	0.422	0.130	0.682	0.318
LR	Test 4 (Peugeot J2)	0.613	0.600	0.056	0.513	0.100	0.617	0.383
SVM	Test 4 (Peugeot J2)	0.650	0.630	0.068	0.578	0.122	0.653	0.347

Test 1 (Opel Hold-out): The SVM model was the clear winner on the hold-out data. As shown in Figure 1, the SVM achieved an AUC of 0.89, significantly higher than the LR's 0.74. By setting the threshold to 0.25, the SVM achieved an excellent Recall of 0.848 for the 'Aggressive' class while maintaining 0.769 Specificity for the 'EvenPace' class. The LR model, with its threshold tuned to 0.45, was forced into a severe trade-off, achieving high Recall (0.825) but at the cost of very low Specificity (0.532). Lower thresholds worsen the performance. This indicates the SVM's non-linear 'rbf' kernel was far more effective at finding a meaningful boundary in the balanced training data.

Test 2 (Opel New Journey): Performance dropped for both models, as seen in Figure 2. This suggests that driving patterns, even for the same driver and vehicle, are not perfectly consistent between journeys. The SVM maintained a higher Accuracy (0.700 vs 0.629) and Specificity (0.749 vs 0.626), but its AUC was surprisingly lower than the LR model's (0.684 vs 0.724).

Tests 3 & 4 (Peugeot): As expected, both models struggled significantly when applied to a different vehicle. The data from the Peugeot was far more imbalanced (over 94% 'EvenPace'), and its sensor characteristics likely differed from the Opel. In both tests (Table 1), the AUC scores for both models hovered between 0.57 and 0.63, indicating a very weak ability to distinguish between classes. However, due to the aggressive thresholding, the SVM consistently achieved a higher Recall for the 'Aggressive' class (0.422 and 0.578) compared to the LR model (0.348 and 0.513). While the precision for both is extremely low (6-8%), the SVM is still the superior model for the stated goal of identifying aggressive instances, even if it generates many false positives.

4 Conclusion

This paper successfully implemented and compared a complete machine learning pipeline for classifying driving styles. The full methodology included feature selection with RFE, class balancing with SMOTE, and hyperparameter tuning with `GridSearchCV`.

The results demonstrate that for this dataset, the tuned Support Vector Machine (with an 'rbf' kernel and $C=5000$) is a significantly more powerful classifier than Logistic Regression, especially on data from the same source (Test 1), where it achieved an impressive 0.89 AUC.

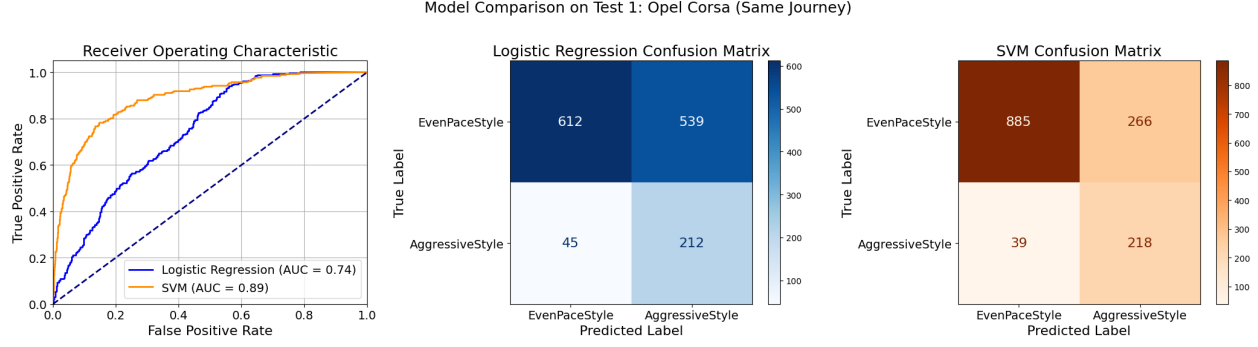


Figure 1: Model Comparison on Test 1: Opel Corsa (Same Journey).

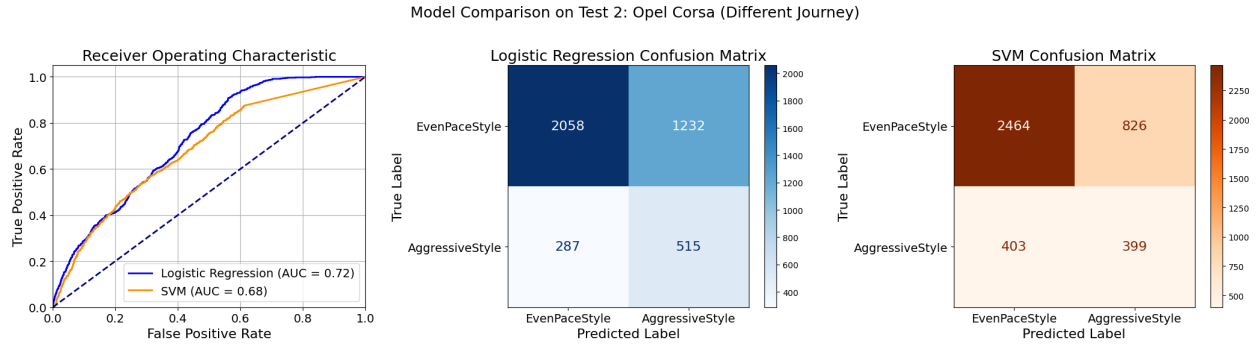


Figure 2: Model Comparison on Test 2: Opel Corsa (Different Journey).

However, the generalization of *both* models to a different vehicle (Peugeot) was poor, with AUCs barely exceeding 0.63. This strongly suggests that the sensor data for aggressive driving is not universal and is highly dependent on the specific vehicle. While the SVM's high recall (achieved via thresholding) makes it more useful for the task of flagging potential aggressive events.

References

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